



## Solve with us 11 - Not Graded



This assignment will not be graded and is only for practice.

Higher order partial derivatives: Higher order partial derivatives:

- Let  $f$  be a scalar valued function defined on a domain  $D$  in  $R^2$ . Then the second order partial derivatives are the partial derivatives of the partial derivatives.

Notations: Notations:

- The partial derivative of  $\frac{\partial f}{\partial x} \frac{\partial}{\partial x} f$  with respect to  $x$  is denoted by  $f_{xx} = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \frac{\partial f}{\partial x}$ .
  - The partial derivative of  $\frac{\partial f}{\partial y} \frac{\partial}{\partial x} f$  with respect to  $x$  is denoted by  $f_{yx} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \frac{\partial f}{\partial y}$ .
  - The partial derivative of  $\frac{\partial f}{\partial x} \frac{\partial}{\partial y} f$  with respect to  $y$  is denoted by  $f_{xy} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \frac{\partial f}{\partial x}$ .
  - The partial derivative of  $\frac{\partial f}{\partial y} \frac{\partial}{\partial y} f$  with respect to  $y$  is denoted by  $f_{yy} = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \frac{\partial f}{\partial y}$ .
- Let  $f(x_1, x_2, \dots, x_n)$  be a function defined on a domain  $D$  in  $R^n$ . Then the second order partial derivatives of  $f$  are defined analogously as the partial derivatives of the partial derivatives.

Notation: Notation: The partial derivative of  $\frac{\partial f}{\partial x_i} \frac{\partial}{\partial x_j} f$  with respect to  $x_j$  is denoted as  $f_{x_i x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$ .

Note: Note: In general, the mixed partial derivatives  $f_{x_i x_j}$  and  $f_{x_j x_i}$  are not necessary equal.

Clairaut's Theorem: Clairaut's Theorem : Let  $f(x_1, x_2, \dots, x_n)$  be a scalar valued function defined on a domain  $D$  containing a point  $a = (a_1, a_2, \dots, a_n)$  in  $R^n$ . If both  $f_{x_i x_j}$  and  $f_{x_j x_i}$  are defined and continuous in an open ball containing the point  $a$ , then

$$f_{x_i x_j}(a) = f_{x_j x_i}(a).$$

$k$ th-order partial derivatives: Let  $f(x_1, x_2, \dots, x_n)$  be a function defined on a domain  $D$  in  $R^n$ . Then the  $k$ th-order partial derivatives of  $f$  are defined analogously by taking successive partial derivatives for  $k$ -times.

Notation: Notation: The notation for  $k$ th-order partial derivative is

$$f_{x_{i_1} x_{i_2} \dots x_{i_k}} = \frac{\partial^k f}{\partial x_{i_k} \dots \partial x_{i_2} \partial x_{i_1}} = \frac{\partial}{\partial x_{i_k}} \dots \frac{\partial}{\partial x_{i_2}} \frac{\partial}{\partial x_{i_1}} f$$

Modified statement of Clairaut's Theorem: Modified statement of Clairaut's Theorem : Let  $f(x_1, x_2, \dots, x_n)$  be a scalar valued function defined on a domain  $D$  containing a point  $a = (a_1, a_2, \dots, a_n)$  in  $R^n$  whose  $k$ th-order partial derivatives are all continuous in an open ball containing the point  $a$ . For any integers  $i_1, i_2, \dots, i_k$  between 1 and  $n$ , if  $j_1, j_2, \dots, j_k$  is a reordering of  $i_1, i_2, \dots, i_k$ , then



$$f_{x_{i_1} x_{i_2} \dots x_{i_k}}(a) = f_{x_{j_1} x_{j_2} \dots x_{j_k}}(a). f_{x_{i_1} x_{i_2} \dots x_{i_k}}(\sim a) = f_{x_{j_1} x_{j_2} \dots x_{j_k}}(\sim a).$$

A consequence of this theorem is that  
we don't need to keep track of the order in which we take partial derivatives.  
we don't need to keep track of the order in which we take partial derivatives.

Hessian matrix: Let  $f(x_1, x_2, \dots, x_n)$  be a scalar valued function defined on a domain  $D$  in  $R^n$ . Then the Hessian matrix of  $f$  is defined as

$$Hf = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \dots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix}. Hf = \begin{pmatrix} \partial_{x_1}^2 f & \partial_{x_1} \partial_{x_2} f & \dots & \partial_{x_1} \partial_{x_n} f \\ \partial_{x_2} \partial_{x_1} f & \partial_{x_2}^2 f & \dots & \partial_{x_2} \partial_{x_n} f \\ \vdots & \vdots & \ddots & \vdots \\ \partial_{x_n} \partial_{x_1} f & \partial_{x_n} \partial_{x_2} f & \dots & \partial_{x_n}^2 f \end{pmatrix}.$$

$Hf(a_1, a_2, \dots, a_n)$  means that all of the entries of  $Hf$  are evaluated at the point  $(a_1, a_2, \dots, a_n)$ , which must be a point in the domain of  $f$ .

1 point

Find  $\frac{\partial^2 f}{\partial x \partial y}$  of the function  $f(x, y) = x^2 y^3 - xy + 6x + 8y + 1$ .

$$f(x, y) = x^2 y^3 - xy + 6x + 8y + 1.$$

☐

$$2x y^3 - y + 62xy^3 - y + 6$$

☐

$$3x^2 y - x + 63x^2 y - x + 6$$

☐

$$6x y^2 - 1$$

☐

00

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$6x y^2 - 1$$

1 point

Find  $\frac{\partial^2 f}{\partial x^2}$  of the function  $f(x, y) = x^5 + xy^7 + e^{xy}$  at point  $(1, -1)$ .

☐

$$20 + e$$

☐

$$20 - e^{-1} 20 - e - 1.$$

☐

$$20 - e 20 - e.$$

☐

$$20 + e^{-1} 20 + e - 1.$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$20 + e^{-1} 20 + e - 1.$$

1 point

Choose the correct statements about  $f(x, y, z) = -x^4 + 2x^2 y^2 z^2 - z^2 y^4 + 2y^2 - xz^4 + 2z^2$ .  
 $f(x, y, z) = -x^4 + 2x^2 y^2 z^2 - z^2 y^4 + 2y^2 - xz^4 + 2z^2$ .

☐

$$\frac{\partial^3 f}{\partial x \partial y \partial z} = 16xyz \quad \partial x \partial y \partial z \partial^3 f = 16xyz$$

☐

$$\frac{\partial^3 f}{\partial z \partial y \partial x} = 16xyz - 8zy^3 \quad \partial z \partial y \partial x \partial^3 f = 16xyz - 8zy^3$$

☐

$$\frac{\partial^3 f}{\partial z \partial y \partial x} = 16xyz \quad \partial z \partial y \partial x \partial^3 f = 16xyz$$

☐

$$\frac{\partial^4 f}{\partial z^4} = -24x \quad \partial z^4 \partial^4 f = -24x$$

☐

$$\frac{\partial^5 f}{\partial y^5} = -24 \quad \partial y^5 \partial^5 f = -24$$

☐

$$\frac{\partial^5 f}{\partial y^5} = -24z^2 \quad \partial y^5 \partial^5 f = -24z^2$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$\frac{\partial^3 f}{\partial x \partial y \partial z} = 16xyz \quad \partial x \partial y \partial z \partial^3 f = 16xyz$$

$$\frac{\partial^3 f}{\partial z \partial y \partial x} = 16xyz \quad \partial z \partial y \partial x \partial^3 f = 16xyz$$

$$\frac{\partial^4 f}{\partial z^4} = -24x \quad \partial z^4 \partial^4 f = -24x$$

1 point

The determinant of the Hessian matrix of  $f(x, y) = x^2 + 3xy + 7y^2$  at point  $(x, y)$  is

☐

2828

☐

1919

☐

00

☐

-19-19

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

1919

1 point

Let  $AA$  denote the Hessian matrix of the function  $f(x, y, z) = 3x^2 + xz + 2zy - z^2$   
 $f(x, y, z) = 3x^2 + xz + 2zy - z^2$  at  $(x, y, z)$ . Which of the following statements is (are) true?



$$A = \begin{pmatrix} 60 & 1 \\ 0 & 2 \\ 12 & -2 \end{pmatrix} A = \begin{pmatrix} \phantom{0} & \phantom{0} \\ 60 & 100212-2 \end{pmatrix}$$



$$A = \begin{pmatrix} -60 & -1 \\ 0 & 2 \\ 1 & -2 \end{pmatrix} A = \begin{pmatrix} \phantom{0} & \phantom{0} \\ -60 & 1002-12-2 \end{pmatrix}$$



$$\text{Det}(A) = 24(A) = 24$$



$$\text{Det}(A) = -24(A) = -24$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$A = \begin{pmatrix} 60 & 1 \\ 0 & 2 \\ 12 & -2 \end{pmatrix} A = \begin{pmatrix} \phantom{0} & \phantom{0} \\ 60 & 100212-2 \end{pmatrix}$$

$$\text{Det}(A) = -24(A) = -24$$

1 point

Let  $AA$  denote the Hessian matrix of the function  $f(x, y, z) = x^2 y - y^3 z^2$   $f(x, y, z) = x^2 y - y^3 z^2$  at  $(1, 1, 1)(1, 1, 1)$ . Which of the following statements is (are) true?



$$\text{Rank}(A) = 3(A) = 3$$



$$\text{Rank}(A) = 2(A) = 2$$



$$\text{Nullity}(A) = 1(A) = 1$$



$$\text{Nullity}(A) = 0(A) = 0$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$\text{Rank}(A) = 3(A) = 3$$

$$\text{Nullity}(A) = 0(A) = 0$$

The Hessian test: Method for finding points of local maxima and minima for a scalar valued function of two  
The Hessian test: Method for finding points of local maxima and minima for a scalar valued function of two

Let  $f(x, y)f(x, y)$  be a function defined on a domain  $DD$  in  $R^2.R^2$ . Suppose that  $f(x, y)f(x, y)$  and its first and second order partial derivatives are continuous in an open ball around  $(a, b) \in D(a, b) \in D$  and  $f_x(a, b) = f_y(a, b) = 0.f_x(a, b) = f_y(a, b) = 0$ . Then

1.  $f$  has a local minimum at  $(a, b)(a, b)$  if  $f_{xx}(a, b) > 0f_{xx}(a, b) > 0$  and  $\det(Hf(a, b)) > 0$   
( $Hf(a, b)) > 0$ .

2.  $f$  has a local maximum at  $(a, b)$  if  $f_{xx}(a, b) < 0$  and  $\det(Hf(a, b)) > 0$ .
3.  $f$  has a saddle point at  $(a, b)$  if  $\det(Hf(a, b)) < 0$ .
4. The test is inconclusive at  $(a, b)$  if  $\det(Hf(a, b)) = 0$ .

1 point

Consider the function  $f(x, y) = x^3 + x^2 y - y$ . Which of the following are correct?

☐

$(-1, \frac{3}{2})$  is a local minimum.

☐

$(-1, \frac{3}{2})$  is a saddle point.

☐

$(1, \frac{3}{2})$  is a local maximum.

☐

$(1, \frac{3}{2})$  is a saddle point.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$(-1, \frac{3}{2})$  is a saddle point.

$(1, \frac{3}{2})$  is a saddle point.

1 point

Choose the correct statement about  $f(x, y) = x^2 - y^3 - x^2 y + y$ .

☐

$(0, \frac{1}{\sqrt{3}})$  is a saddle point.

☐

$(0, \frac{1}{\sqrt{3}})$  is a saddle point.

☐

$(0, \frac{-1}{\sqrt{3}})$  is a local minimum.

☐

$(0, \frac{-1}{\sqrt{3}})$  is a local minimum.

☐

$(0, \frac{-1}{\sqrt{3}})$  is a local maximum.

☐

$(0, \frac{-1}{\sqrt{3}})$  is a local maximum.

☐

Hessian test is inconclusive at the point  $(0, \frac{1}{\sqrt{3}})$ .

☐

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$(0, \frac{1}{\sqrt{3}})$  is a saddle point.

☐

$(0, \frac{1}{\sqrt{3}})$  is a saddle point.

☐

$(0, \frac{-1}{\sqrt{3}})$  is a local minimum.

☐

$(0, \frac{-1}{\sqrt{3}})$  is a local minimum.

1 point

Choose the correct statement about  $f(x, y) = 2x^3 + 6xy^2 - 3y^3 - 150x$ .

$$f(x, y) = 2x^3 + 6xy^2 - 3y^3 - 150x.$$

☐

$(5, 0)$  is a local minimum.

☐

$(-3, -4)$  is a saddle point.

☐

$(-5, 0)$  is a local minimum.

☐

$(-5, 0)$  is a saddle point.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$(5, 0)$  is a local minimum.

$(-3, -4)$  is a saddle point.

The Hessian test: Method for finding points of local maxima and minima for a scalar valued function of three variables.

Let  $f(x, y, z)$  be a function defined on a domain  $D$  in  $\mathbb{R}^3$ . Suppose that  $f(x, y, z)$

and its first and second order partial derivatives are continuous in an open ball around

$(a, b, c) \in D$  and  $f_x(a, b, c) = f_y(a, b, c) = f_z(a, b, c) = 0$ .

Then

- $f$  has a local minimum at  $(a, b, c)$  if  $f_{xx}(a, b, c) > 0$ ,  $(f_{xx}f_{yy} - f_{xy}^2)(a, b, c) > 0$ , and  $\det(Hf(a, b, c)) > 0$ .
- $f$  has a local maximum at  $(a, b, c)$  if  $f_{xx}(a, b, c) < 0$ ,  $(f_{xx}f_{yy} - f_{xy}^2)(a, b, c) > 0$ , and  $\det(Hf(a, b, c)) < 0$ .
- $f$  has a saddle point at  $(a, b, c)$  if  $\det(Hf(a, b, c)) \neq 0$  and cases 1 and 2 do not occur.
- The test is inconclusive at  $(a, b, c)$  if  $\det(Hf(a, b, c)) = 0$ .

1 point

Consider the function  $f(x, y, z) = x^3 + xz^2 - 3x^2 + y^2 + 2z^2$

$f(x, y, z) = x^3 + xz^2 - 3x^2 + y^2 + 2z^2$ . Which of the following options are correct?

☐

$(0, 0, 0)$  is a local maximum.

☐

$(0, 0, 0)$  is a saddle point.

☐

Hessian test is inconclusive at the point  $(0, 0, 0)$ .

☐

$(0, 0, 0)$  is a local maximum.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$(0, 0, 0)$  is a saddle point.

1 point

Choose the correct statements about  $f(x, y, z) = e^x(x^2 - y^2 - 2z^2)$

☐

$(0, 0, 0)$  is a local maximum.

☐

$(0, 0, 0)$  is a saddle point.

☐

Hessian test is inconclusive at the point  $(-2, 0, 0)$ .

☐

$(-2, 0, 0)$  is a local maximum.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$(0, 0, 0)$  is a saddle point.

$(-2, 0, 0)$  is a local maximum.

Differentiability for scalar valued multivariable function:

Differentiability for scalar valued multivariable function:

Let  $f(x_1, x_2, \dots, x_n)$  be a scalar valued function defined on a domain  $D$  in  $\mathbb{R}^n$  containing an open ball around a point  $a = (a_1, a_2, \dots, a_n)$ . Then  $f$  is differentiable at  $a$  if

$$\lim_{h \rightarrow 0} \frac{f(a + h) - f(a) - h \cdot \nabla f(a)}{\|h\|} = 0$$

$$\|f(a + h) - f(a) - h \cdot \nabla f(a)\| = 0$$

1 point

Which of the following functions are differentiable at  $(0, 0)$ ?

☐

$$f(x, y) = x^2 + y^2$$

☐

$$f(x, y) = (xy)^{\frac{1}{3}}$$

☐

$$f(x, y) = \begin{cases} \frac{x|y|}{\sqrt{x^2 + y^2}} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

☐

$$f(x, y) = \begin{cases} x|y| & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

☐

$$f(x, y) = e^{x^2 + y^2}$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$f(x, y) = x^2 + y^2$$

$$f(x, y) = e^{x^2 + y^2}$$

[Check Answers](#)

Your score is: 0/12